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Observed Northward Migration of Agro-Climate Zones in Europe Will Further Accelerate Under Climate Change

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Key Points:
- Agro-climate zones in eastern Europe have experienced a northward migration velocity of 100 km per 10 years over the past 40 years
- Northward migration of agro-climate zones in Europe may be up to two times faster in next decades
- Negative impacts of heat stress are expected to nonlinearly increase in large parts of southern and southeastern Europe

Supporting Information:
- Supporting Information S1
- Figure S1

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Abstract This study focuses on the northward shift of homogeneous agro-climate zones in Europe analyzed for the observed past and projected climate conditions for the next decades. Statistical cluster analysis is used to derive eight main agro-climatic zones driven by two agro-meteorological indicators, namely, active temperature sum and thermal growing season length. The northward shift of homogeneous agro-climate zones and the corresponding change of crop growth suitability are analyzed together with the change of exposure of crops to temperature-related climate extremes during the growing season. Gradual warming over Europe has contributed to a lengthening of the growing season and an increased active temperature accumulation, accompanied by more frequent occurrence of warm extreme climate events. Using a set of five high-resolution regional climate scenarios, we calculate that a major part of Europe will be affected by further northward climate zone migration. In the next decades, the migration of agro-climatic zones in Eastern Europe may reach twice the velocity observed during the period 1975–2016. Several regions of the Mediterranean may lose suitability to grow specific crops in favor of northern European regions. This indicator-based assessment suggests that the potential advantages of the lengthening of the thermal growing season in northern and eastern Europe are often outbalanced by the risk of late frost and increased risk of early spring and summer heat waves.

1. Introduction

Temperature is the major abiotic factor affecting the phenological development of plant species, with temperatures accumulated over time (i.e., thermal time) characterizing physiological development of plants throughout their life cycle (Trudgill et al., 2005). Although thermal requirements for growth differ among plants species, the observed climate warming has already contributed significantly to earlier occurrence of phenophases of common plant species, including herbs, agricultural crops, fruit trees, and forests (Schleip et al., 2009), with important consequences for plant health and biodiversity. For existing agricultural crop production systems, the crop cycle has shortened due to increasing temperatures in recent decades (Fraga et al., 2012; Olesen et al., 2011, 2017).

There is mounting evidence that the warming has already strongly modified agricultural crop production systems (Olesen et al., 2011, 2017; Schleip et al., 2009) as well as ecosystems location and composition in Europe (Pautasso, 2012), beyond phenological development stages. Extreme drought and heat stress conditions had detrimental impacts on agricultural crop production in southern Europe, where moderate to extreme problems have been experienced (Fontana et al., 2015; Olesen et al., 2011; Zampieri et al., 2017). The observed warming has also promoted the shift of crop species to areas, which were previously constrained by either a too short growing season length (*GSL*) or by unreachable thermal requirements to finish the crop growth cycle (Hannah et al., 2013; Marx et al., 2017; Olesen et al., 2011, 2017; Park et al., 2016). A well-documented example is the shift in the climatic suitability for wine production, which was spatially mostly confined to Mediterranean, southern maritime, and Pannonian regions in Europe (Hannah et al., 2013), and which expanded in the recent period in several regions of central and western Europe (Spinoni et al., 2015). In addition to more frequent and intense heat waves and droughts, anomalous wet conditions have often significantly contributed to crop yield reduction (Ben-Ari et al., 2018; Brisson et al., 2010; Ceglar et al., 2018; Olesen et al., 2017; Trnka et al., 2015).

Expected climate change will likely promote further northward expansion of crops adapted to warmer climates but also decrease suitability in the areas affected by increasingly higher temperatures and more frequent droughts (Hannah et al., 2013; Iglesias et al., 2011; King et al., 2018; Marx et al., 2017;

Peltonen-Saino, 2009). King et al. (2018) suggested that crop suitable growing-degree-days conditions will experience a northward expansion of up to 1,200 km in the boreal regions by the end of the 21st century. However, northward expansion of conditions conducive to agricultural crop production in the boreal regions will be accompanied by the migration of agro-climate zones and crop growing conditions also elsewhere in Europe.

The migration velocity of homogeneous climate zones can be considered as a direct measure of varying temperature-driven suitability for agriculture and ecosystems. This principle has previously been used as an indicator for climate change threats to ecosystems (Loarie et al., 2009). Although ecological barriers (such as mountains and lakes) can temporally halt migration, the migration velocity of climate zone borders is substantially affecting the spatial redistribution of ecological communities (Waldock et al., 2018). Many plant and animal species can therefore struggle to remain in favorable temperature zones (Pearson, 2006). As ecological barriers are less important for agricultural production systems, which are mainly located in flat areas and actively managed, the migration of agro-climatic zones and suitability for growing specific crops is likely to take place at a higher speed than those for natural ecosystems composition.

This study aims to explore (*i*) the implications of migration of climate zones in Europe for agricultural crop production, (*ii*) identify potential beneficial and detrimental effects in various regions, and (*iii*) assess additional risks from temperature-driven extreme climate events. We quantify the observed and future velocity of northward migration of European agro-climate zones, focusing on suitability for crop production. For this purpose, we develop a dedicated climate zonation based on *GSL* and active temperature sum (*ATS*), thereby characterizing the potential for crop production under specific climate conditions. Along with changes of average climate conditions (expressed by *GSL* and *ATS*), also, extreme climatic conditions may change, often in a nonlinear way. This implies that while growth conditions may be well characterized by the climate zone, crop damaging conditions such as extreme heat stress may change at different rates for that climate. To quantify this effect, we additionally use an indicator-based approach to assess the area affected by different temperature extremes across Europe and discuss implications of past and current observed and projected future temperature changes for crop yields of the main European crop (winter wheat).

2. Data and Methods

Observed daily minimum and maximum air temperatures were obtained from the MARS Crop Yield Forecasting System (MCYFS) database, established and maintained by the Joint Research Centre of the European Commission for the purpose of crop growth monitoring and forecasting (Biavetti et al., 2014). In short, daily meteorological data are obtained from around 4,200 weather stations, quality controlled, gap-filled, and interpolated into a regular 25 × 25 km grid over Europe and neighboring countries. The advantage of using MCYFS, compared to the more widely used E-OBS (Cornes et al., 2018) or ERA-interim (Dee et al., 2011) databases, is given by the higher spatial density of quality-controlled meteorological station data in European agricultural areas, which are used for the interpolation of five meteorological variables (daily precipitation, downwelling shortwave radiation, 2-m air temperature, relative humidity/partial water vapor pressure, and wind speed), covering the period from 1975 up to near real time. Since we have a large user experience with MCYFS for crop modeling and monitoring on European level (e.g., Toreti, Maiorano, et al., 2019), MCYFS is a safe choice for this study as well.

Future climate projections were retrieved from a set of high-resolution bias-corrected EURO-CORDEX regional climate models (Dosio, 2016). High-end emission scenario (RCP8.5) has been considered here. Even though EURO-CORDEX provides an ensemble of 11 different climate model simulations, our study selects five models, encompassing the full range of results produced by the larger set of 11 models (Dosio, 2016). Table 1 shows the selected models and the future periods of focus for this study.

While climate variables such as monthly and seasonal temperature and precipitation have been often used to define climate zones across Europe (e.g., Beck et al., 2018; Metzger et al., 2005), we compute climate zonation on *GSL* and *ATS*, two variables defining a measure of the climate-driven potential of agricultural production systems. Even though the growing season characteristics are dependent on crop species, we follow here the generic recommendation from the World Meteorological Organization (Klein-Tank et al., 2009), which defines the *GSL* as the length of the period between first date when daily mean temperature exceeds 5 ° C for five consecutive days and first date when mean daily mean temperature drops below 5 ° C for five consecutive days. The *ATS* is a canonical metric to describe the consequent plant phenological

Table 1
Climate Models From the High-Resolution EURO-CORDEX Simulations, Selected for This Study

Acronym	Institute	RCM	GCM	+2 ° C
RCM1	CLMcom	CCLM4-8-17	ICHEC-ECEARTH	2027–2056
RCM2	CLMcom	CCLM4-8-17	CNRM-CERFACS-CNRM-CM5	2033–2062
RCM3	IPSL-INERIS	WRF331F	IPSL-IPSL-CM5A-MR	2021–2050
RCM4	SMHI	RCA4	MOHC-HadGEM2-ES	2016–2045
RCM5	SMHI	RCA4	MPI-M-MPI-ESM-LR	2030–2059

Note. More details on EURO-CORDEX simulations are provided in Jacob et al. (2013). The last column (+2 ° C) indicates for the RCP8.5 scenario the 30-year period when an average global temperature increase of 2 ° C is reached compared to preindustrial period.

development stages within the growing season and is calculated by integrating daily temperature above a threshold 0 ° C (the base temperature for winter wheat development) over time. This approach assumes that the temperature and development rate relationship in plants is linear, which is often the case up to the thermal optimum temperature (Trudgill et al., 2005).

K-means clustering, an objective statistical method, has been used to obtain spatial climate zonation of Europe's current climate, with gridded interannual averages of *GSL* and *ATS* as factors. The algorithm resulted in eight clusters (i.e., homogeneous agro-climate zones; Figure S1 in the supporting information), represented by unique pairs of *GSL* and *ATS* (i.e., centroids), corresponding to the average values assigned to each cluster (see the supporting information for more details). This methodology can be consistently used to identify temperature-driven spatial migration velocity of agro-climate zones between different periods.

The observed migration of homogeneous agro-climatic zones is diagnosed by comparing their spatial distribution in two distinct periods: 1975–1995 and 1996–2016. The former period is used to determine the centroids of the reference zones. The same centroids are then used in the period 1996–2016 to determine their spatial migration. For this purpose, the gridded averages of *GSL* and *ATS* in the latter period are assigned to agro-climate zones based on distance with reference zonal centroids:

$$dist_i^k = \sqrt{\left(GSL_i^{STD} - GSL_k^{STD}\right)^2 + \left(ATS_i^{STD} - ATS_k^{STD}\right)^2}, \tag{1}$$

where GSL_i (ATS_i) represents average *GSL* (*ATS*) for *i*th grid point for the period 1996–2016 and GSL_k (ATS_k) represents the reference centroid values for *k*th agro-climate zone, determined for the period 1975–1995. *STD* indicates the standardization of variables. The *i*th grid is assigned to agro-climate zone with minimum distance. The projected migration of (climate model specific) homogeneous agro-climatic zones is calculated using the same procedure, however, taking the same reference period as provided by the climate scenario results (1981–2010). Obtained centroids are then used to calculate migration of zones under a 2 ° C warming scenario (Table 1).

The latitudinal migration velocity is calculated by first taking the latitude of border grid cells of each agro-climate zone in two distinct periods and then calculating the ratio between average latitudinal displacement of border cells and time, in which the displacement occurred.

To better understand temperature-driven crop production limitations, four temperature-driven indicators are analyzed for each agro-climate zone:

- frost days with minimum daily temperature below 0 ° C;
- tropical nights, when minimum daily temperatures exceeds 20 ° C for at least two consecutive days;
- mild heat stress, when maximum daily temperatures exceed 31 ° C for at least two consecutive days; and
- strong heat stress, when maximum daily temperatures exceed 35 ° C for at least three consecutive days.

The selected temperature thresholds are tailored toward characterizing crop growing conditions for winter wheat (i.e., Olesen et al., 2011; Porter & Semenov, 2005; Trnka et al., 2014). Table S2 provides more details on related impacts on crop growth processes. In summary, 0 ° C represents a lower threshold for assessment of exposure of plants to damaging frost, if it occurs after the onset of vegetation period. Additionally, many physiological processes in plants cease below this temperature. Various experiments for winter wheat have shown that maximum temperatures above 31 ° C around the flowering period can reduce the number of

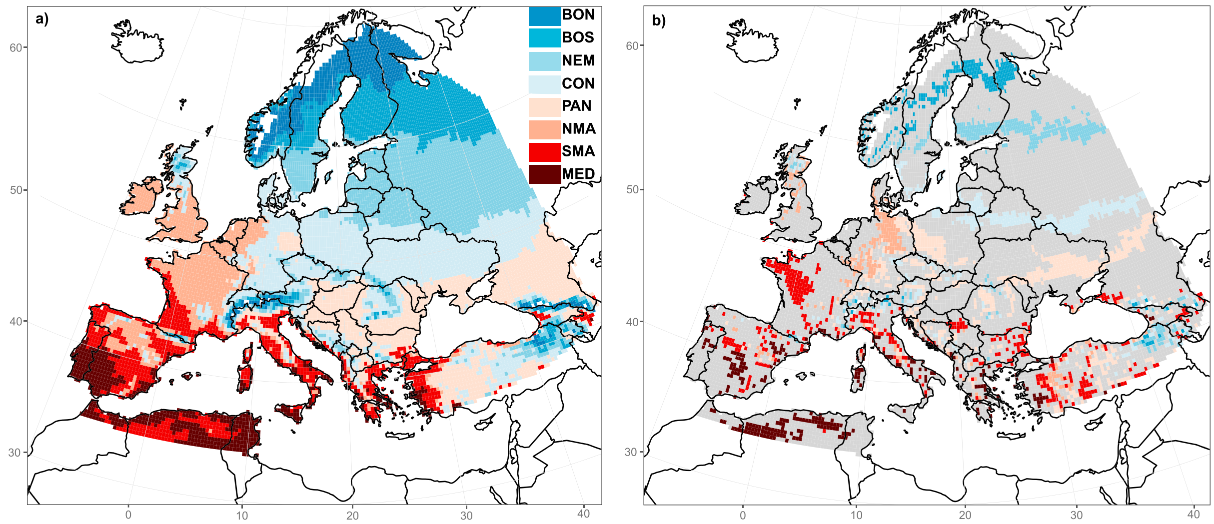

Figure 1. (a) Agro-climate zonation of Europe based on growing season length (*GSL*) and active temperature sum (*ATS*) for the period between 1975 and 1995. The identified agro-climate zones are named as follows (going from north to south): boreal north (*BON*), boreal south (*BOS*), nemoral (*NEM*), continental (*CON*), Pannonian (*PAN*), northern maritime (*NMA*), southern maritime (*SMA*) and Mediterranean (*MED*). (b) The migration of agro-climate zones between the 1975–1995 and 1996–2016 periods. For better distinction, only the areas affected by migration of agro-climate zones are displayed (colored areas), while gray color denotes the areas where the agro-climate zones have not changed.

grains (Rezaei et al., 2018), while temperatures above 35 ° C affect the development of the endosperm, which limits maximum grain weight during the grain filling stage (Hawker & Jenner, 1993). Finally, the number of tropical nights, when minimum temperatures fail to fall below 20 ° C, is a well-established risk factor for human health. Although the evidence for impacts on plants is less, increasing tropical nights may also affect nighttime plant respiration rates, leading to reduced biomass accumulation and crop yield (Hatfield & Prueger, 2015).

Above-mentioned agro-climate indicators are calculated for each agro-climate zone, followed by an estimation of the area affected in given calendar day or during thermal growing period, specified dynamically (i.e., in terms of phenological timing) by *ATS*.

3. Observed Shifts in European Agro-Climate Zones

The clustering method resulted in eight homogeneous agro-climate zones (Figures 1 and S3). In eastern Europe, these are characterized by a coherent latitudinal bands. In western and southern Europe, the classification results in more complex spatial pattern, reflecting the influence of Atlantic ocean, the Mediterranean Sea, and complex orography. Generally, the identified agro-climate zones correspond well to environmental stratification of Metzger et al. (2005) and Köppen-Geiger ($K-G$, Beck et al., 2018), indicating the leading role of the temperature in spatial distribution of environmental zones in Europe (Mahlstein et al., 2013; Zampieri & Lionello, 2010). Nevertheless, several noticeable differences can be observed when compared with $K-G$ classification. In the latter, prevalence of temperate continental climate characterizes eastern Europe, where (instead of one) two zones have been identified by clustering method used in this study. Additionally, differences can be observed also in southeastern Europe (especially the Pannonian Basin) and western Europe. Our results distinguish between growing season characteristics of southwestern and northern France, which is not the case in $K-G$ classification with prevalence of temperate oceanic climate over the whole country.

Observed differences are less obvious when compared to environmental stratification of Metzger et al. (2005). To avoid technical jargon in referring to obtained agro-climate zones, and given their similarity in spatial distribution with Metzger et al. (2005), we use already established nomenclature for their naming (Figure 1), despite minor differences in the exact definitions of these zones in our method.

The observed shift of homogeneous agro-climate zones is diagnosed by comparing their spatial extent in two distinct periods: 1975–1995 and 1996–2016. The former period is used as a reference to determine the centroids of the eight agro-climatic zones. In this period, the *GSL* roughly spans from 128 days in the northern boreal zone to 363 days in the Mediterranean. *ATS* spans from 1,300 degree days (DD) in the northern boreal

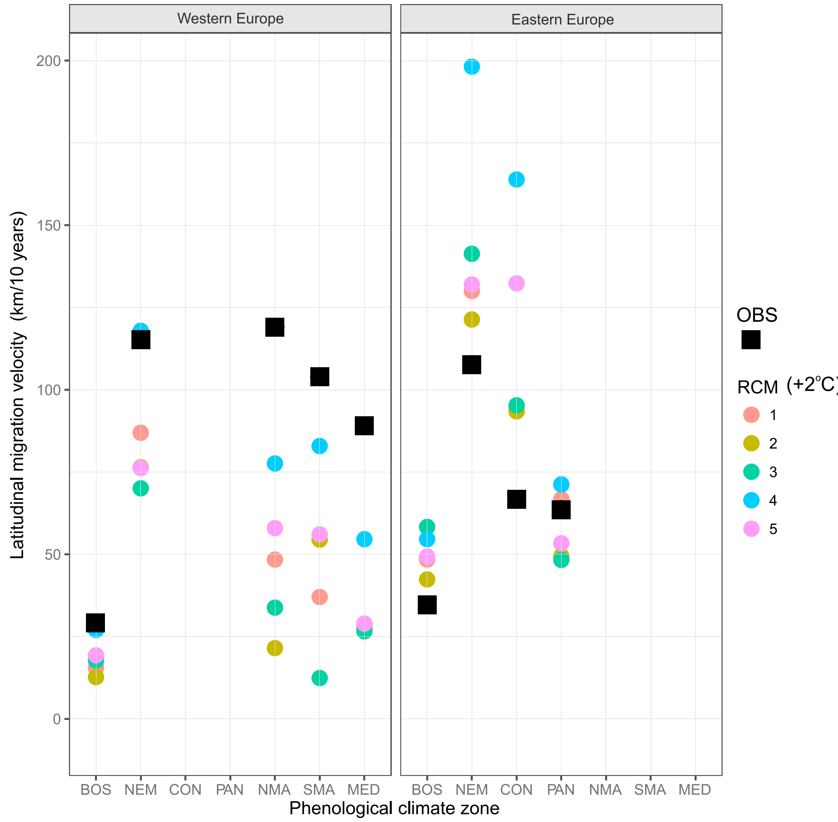

Figure 2. Latitudinal migration velocity of agro-climate zone borders, calculated separately for the western and eastern Europe (border defined by the 15° E meridian). For each agro-climate zone, the migration velocity is calculated as an average rate of change of latitude of zonal borders, normalized to 10-year period. Observed speed (black dots) is calculated based on comparison of the average border location between 1975–1995 and 1996–2016. Considering climate projections, migration velocities are shown between the reference period 1981 and 2010 and the future period, which is model specific (corresponding to 2° C global warming level; Table 1).

zone to almost 6,200 DD in the Mediterranean (Table S1). The same centroids are then used for the entire period 1996–2016 to determine the spatial migration of the reference agro-climate zones (see section 2).

Comparing 1996–2016 to 1975–1995, Europe experienced a considerable migration of agro-climatic zones toward the North with respect to the earlier period (Figure 1). The migration velocity has been calculated for prevailing agro-climate zones in western and eastern Europe (bordered by 15° meridian) separately. In eastern Europe, the migration in recent decades has been most pronounced for the nemoral zone, with an estimated latitudinal velocity of the zonal border toward the north exceeding 100 km per 10 years (Figure 2). The continental and Pannonian zones have been progressing northward at almost 70 km per 10 years. Half of this velocity (35 km per 10 years) is estimated for southern boreal zone. Also, western Europe experienced a pronounced northward migration of maritime and Mediterranean climate zones. There has been a substantial shift of the northern maritime zone toward the northeastern direction, appearing in regions such as central-northern Germany, which typically had a continental climate during the 1975–1995 period. A striking northward shift has also affected the southern maritime climate in France, which was regionally confined to southwestern and southern parts of the country during the reference period. Similarly, it has expanded also in the Po Valley of northern Italy, where growing season characteristics of the colder Pannonian climate regime could be found during the reference period.

The northward migration of the nemoral into the formerly southern boreal zone is among the most prominent features in the observed changes during the observational period. Observed migration of agro-climate zones in large parts of Europe (most pronouncedly in southeastern and northern Europe) is coinciding with a large positive trend in thermal growing season length, the earlier season onset, and delay of end of season

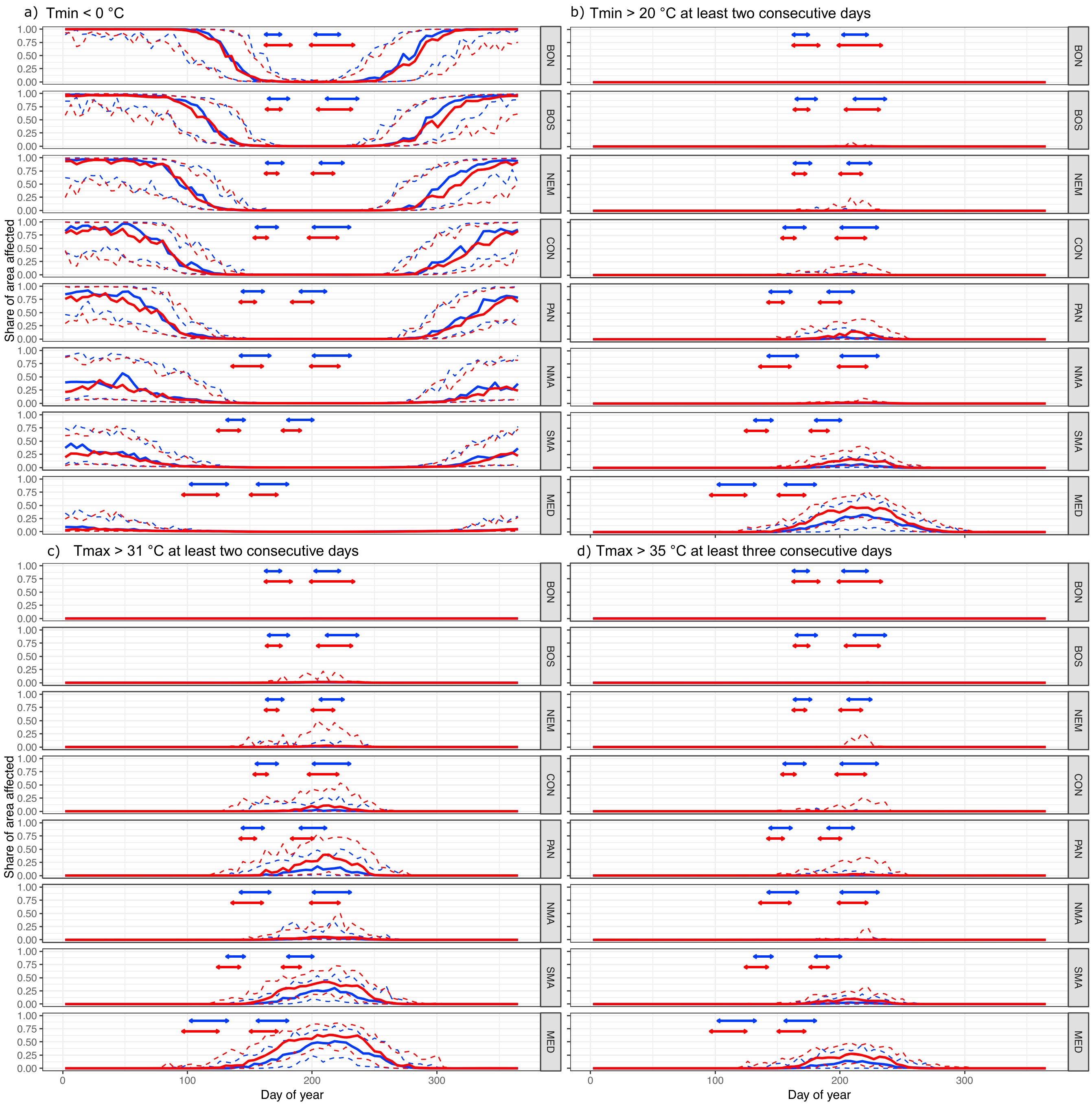

Figure 3. Relative area (fraction of the whole agro-climate zone) affected by climatic events based on four temperature-driven indicators for different agro-climate zones: minimum temperature below 0 ° C (a), at least two consecutive days with minimum temperature above 20 ° C (b), at least two consecutive days with maximum temperature above 31 ° C (c), and at least three consecutive days with maximum temperature above 35 ° C (d). Thick lines represent median of the area affected (estimated as 20-year median for each day of year), whereas the dashed lines represent interannual range of affected areas on a given day of year. Blue color refers to the period between 1975 and 1995, whereas red color refers to the period between 1996 and 2016. Arrows represent the interquartile range (i.e., spread between 25th and 75th percentile) of simulated flowering (first pair of arrows on *x* axis) and maturity dates (second pair of arrows on *x* axis) for winter wheat within each agro-climate zone, with colors corresponding to the above-mentioned periods. In the case of indicators (b), (c), and (d), the day of year on *x* axis represents the first of the consecutive days. The time series of affected area have been additionally smoothed using LOESS (Cleveland, 1979), considering 10-day influential period.

over the 1975–2016 period (Figure S2). The longer growing season has been shown to contribute to increased vegetation productivity in the boreal regions (Park et al., 2016).

In addition to the migration of climate zones, we analyze changes in the total area affected by climate events, identified by four agro-climate indicators (see section 2), and its coincidence with sensitive phenological stages of winter wheat (Figure 3). The affected area in each zone is calculated separately for two periods: 1975–1995 and 1996–2016. However, for this purpose the spatial extent of agro-climate zones is derived over the entire observational period (1975–2016), as we want to emphasize the temporal changes spatially confined to the same areas (Figure S3). Flowering and maturity dates in the two distinct 20-year periods are simulated by the WOFOST crop model, which has been calibrated to represent spatial distribution of European varieties (Ceglar et al., 2019).

The frost-affected area during the cold half of year has decreased within all agro-climate zones and in particular in eastern Europe in autumn and early winter. In late spring (i.e., second half of April and beginning of May), the share of area affected by frost days has not changed substantially, indicating higher exposure of plants to late spring frost due to earlier growing season onset (Figures 3 and S2). Consequently, the risk of vegetation exposure to frost kill after the vegetation season onset has likely increased, consistent with Liu et al. (2018).

The area affected by at least two consecutive tropical nights has increased during the more recent period within all agro-climate zones, except boreal ones (Figure 3b). The highest increase can be observed in the Pannonian, southern maritime, and Mediterranean zones, similar to the area affected by mild heat stress. Besides the apparent increase during midsummer period, earlier first occurrence in the season can be observed, most apparently in the Mediterranean.

The area affected by mild heat stress (i.e., maximum temperatures above 31 ° C for two consecutive days or more) during late spring, coinciding with winter wheat flowering period, has increased in the Pannonian, continental, southern maritime, and Mediterranean climate zones (Figure 3c). As for the summer season, the area affected by mild heat stress has increased in all climates spanning through central and southern Europe. The area has doubled in early summer in the Mediterranean and southern maritime zones, while the highest increase can be observed in the Pannonian agro-climate zone. The occurrence of mild heat stress events in the southern boreal zone was absent during the 1975–1995 period but has appeared in the subsequent period.

The area affected by strong heat stress (three or more consecutive days with maximum temperature above 35 ° C), which has a detrimental effect on winter wheat during the grain filling period, has been mainly limited to the Mediterranean and southern maritime climates during the reference period (Figure 3d). These events have affected larger areas in more recent period also in the Pannonian, continental and nemoral climates, most pronounced in years 2007, 2010, and 2012 (Ceglar et al., 2018). The area subject to strong heat stress has roughly doubled in the Mediterranean and southern maritime climates.

Earlier flowering and maturity dates of winter wheat are simulated across all European zones in the 1996–2016 with respect to the preceding period. In the boreal agro-climatic regions an increase in *ATS* reflects a larger area suitable for wheat production (Peltonen-Saino, 2009). However, in most European regions, faster temperature accumulation also results in earlier flowering and maturity. Even though the area affected by mild heat stress in late spring (around flowering of winter wheat) has also increased, the earlier occurrence of flowering has partially compensated for negative heat stress effect, especially in continental and nemoral zones. This confirms the findings of Rezaei et al. (2015) for winter wheat grown in Germany, where an earlier date of heading and corresponding shift of the flowering period to the cooler spring season almost completely compensated for the warming trend and heat stress around flowering. However, this is not the case for the Pannonian agro-climate zone, where a doubling of the share of area affected by heat stress is observed despite earlier flowering occurrence. Indeed, wheat grown in this agro-climate zone is more sensitive to heat stress during the flowering and grain filling than in other regions of Europe (Lecerf et al., 2018; Webber et al., 2018). Furthermore, together with the increased area affected, the crop model simulations also indicate shorter periods of flowering time. Shortened duration of flowering in the presence of mild heat stress can further increase yield losses due to shorter time for physiological escape from effects of heat stress (Jones et al., 2016, and references therein). Earlier occurrence of heat stress is observed also for southern maritime and Mediterranean climates, with noticeable effect on durum wheat

(Fontana et al., 2015). The continental, Pannonian, southern maritime, and Mediterranean climates have experienced a substantial increase in area affected by heat stress during the grain filling period (shown by all heat stress indicators in Figures 3b–3d), which has together with regional intensification of drought conditions and shortening of grain filling period contributed to wheat yield stagnation after the 1990s induced by changing climatic conditions (Brisson et al., 2010).

The observed migration of agro-climate zones together with increased frequency and intensity of extreme climatic events has likely contributed to changes in agricultural harvested area across Europe (Olesen et al., 2017). The already observed changes in temperature and precipitation could have contributed to increasing (decreasing) harvested wheat areas in northern (southern) Europe (Figure S5). While changing climate is likely codetermining the European land use patterns, we emphasize that climate change is acting on a background of socioeconomic and political factors that have substantially contributed to changes in crop harvested areas in Europe during the last 30 years. The breakdown of socialism in 1989 in Eastern Europe has been recognized as key triggering event for agricultural abandonment and decline of capital intensive farming practices (Kuemmerle et al., 2016). During this time, western Europe has benefited from the funding mechanisms under EU's Common Agricultural Policy (CAP), although the extent to which CAP has triggered or prevented land use change is not well established. Nevertheless, the observed increase in intensity and frequency of climatic extremes has already exposed the need for effective policy guidance in the presence of agro-climate zone migration in order to improve resilience of cropping systems to future climate (Spinoni et al., 2018; Toreti, Belward, et al., 2019).

4. Accelerated Northward Migration of European Climates Under Future Climate Change

A shift of European agro-climate zones is further analyzed for five EURO-CORDEX regional climate model runs based on the high-end emission scenario (RCP85) focusing on the 30-year period with 2 ° C of global climate warming (compared to a preindustrial period). This period has been determined based on the mean global warming signal from the driving global climate models. The location and areal extent of the agro-climate zones for temperature simulated by different regional climate models are derived by using the same clustering procedure of the observational analysis.

The agro-climate zones, as simulated by climate models for the baseline period, may have a different geographical extent and do not exactly correspond to the values explained in Table S1. Consequently, different combinations of *GSL* and *ATS* for the same zone can result from the clustering procedure. Nevertheless, climate zonation during the baseline period (1981–2010) generally resulted in *ATS* and *GSL* centroids similar to the ones identified with observed temperature for the period 1975–2016 (Figure S4). An interesting divergence of agro-climatically relevant temperatures can be observed for Pannonian and northern maritime climates. While the observed *ATS* is higher in the former, the observed *GSL* is longer in the latter region. In general, climate models tend to exhibit the largest spread of *ATS* over these two regions, but they agree better on the *GSL*. Given that all climate simulations have been bias corrected, the calculated spread of *ATS* is likely related to differences in simulated internal variability of regional climate model dynamics.

Under a 2 ° C warming scenario according to RCP8.5, all climate model simulations agree on further northward migration of agro-climate zones in Europe (Figures 2 and 4). The nemoral zone is expected to progress roughly 20–100% faster than the past 40 years (Figure 2), taking into consideration the spread between different climate models. This indicates that current southern boreal zone will experience an extension of growing season length of 22 days, while *ATS* during the growing season will increase by around 400 DD, considering the ensemble mean of different climate models. The continental agro-climate zone is expected to migrate with a velocity between 90 and 165 km per 10 years toward the north, which is between 1.4 and 2.5 times faster than the current migration speed. The area under current nemoral climate zone will therefore experience an increase of *GSL* of nearly 20 days and *ATS* increase of 400 DD. Noticeable expansion is foreseen also for the Pannonian zone, especially toward central Europe, where continental climate generally prevailed during the baseline period. Complex patterns of change are predicted for southeastern Europe, generally with a complete agro-climate zone shift for the entire region, with exception of Hungary. The Pannonian climate, which prevailed in central and western Balkans during the baseline period, is expected to diminish from the region. The appearance of southern maritime growing season characteristics dominates in central and western Balkans, indicating lengthening of the growing season for 20 days and increase of *ATS* for 500

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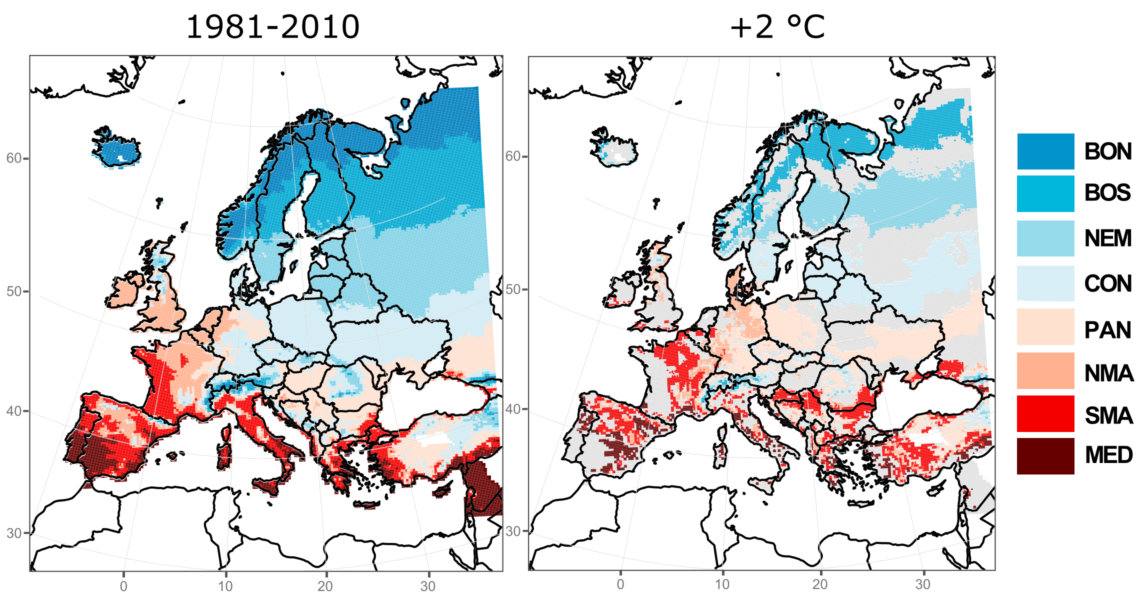

Figure 4. (a) Agro-climate zonation of Europe based on growing season length (*GSL*) and active temperature sum (*ATS*) obtained as an ensemble median from five different CORDEX climate model simulations (Table 1) during the baseline period (1981–2010). (b) Ensemble median spatial patterns of agro-climate zones migration under 2 ° C global surface warming according to RCP8.5 (Table 1). Gray areas represent regions where no change with respect to the baseline period is simulated.

DD, considering the ensemble mean. Another prominent feature is the expansion of Mediterranean zone in Greece, western Turkey, and along the Adriatic coast.

The projected latitudinal migration of agro-climate zones in western Europe is highly diverse among models (Figures 4 and S6). Even though all models agree on the northward direction of velocity (Figure 2), a distinct pattern of eastward expansion of maritime agro-climate zones can also be observed (Figure S6). The simulated eastward longitudinal velocities range between 70 and 170 km per 10 years (5 and 80 km per 10 years) for northern (southern) maritime zone, depending on the climate model. The northern maritime zone is projected to prevail in northern and western Germany, substituting continental regime during the baseline period. The expansion of southern maritime climate is prominent in central and northern France, substituting the northern maritime climate regime. In terms of growing season characteristics, this indicates lengthening of nearly 20 days and *ATS* increase of up to 700 DD. The Mediterranean climate is projected to expand in central and southern Italy, southern France, and eastern part of the Iberian Peninsula, regions which were dominated by the southern maritime growing season characteristics.

An analysis the area affected by climate events defined by different temperature thresholds (see section 2) is now carried out by replacing time expressed in calendar days with phenological timing (i.e., based on active temperature accumulation). This dynamic approach has two main advantages with respect to the conventional static approach: (a) The progress of growing season is measured in terms of temperature accumulation, which can be therefore directly related to crop phenological development, and (b) the impact of extreme climatic events on crops can be aligned with the occurrence of crop development stages (phenological timing) rather than the fixed calendar period. The latter point is especially important for assessing the varying susceptibility of crops to extreme climate events during different phenological stages.

Figure 5 shows the area affected by the four climatic events during different phenological stages for the reference (1981–2010) and model specific future period defined by 2 ° C global warming. For each climate model, daily gridded time series of ATS are calculated for both reference and future periods, indicating the active temperature sum as the season progresses. Then, the ATS is binned in classes of width equal to 100 DD, spanning from 0 to 7,000 DD (i.e., 70 bins). For each ATS bin, the interannual average, minimum and maximum areas affected by the four climatic events are then calculated separately for the reference and scenario periods. This procedure is repeated for all the five climate models (Table 1). Figure 5 summarizes all these information.

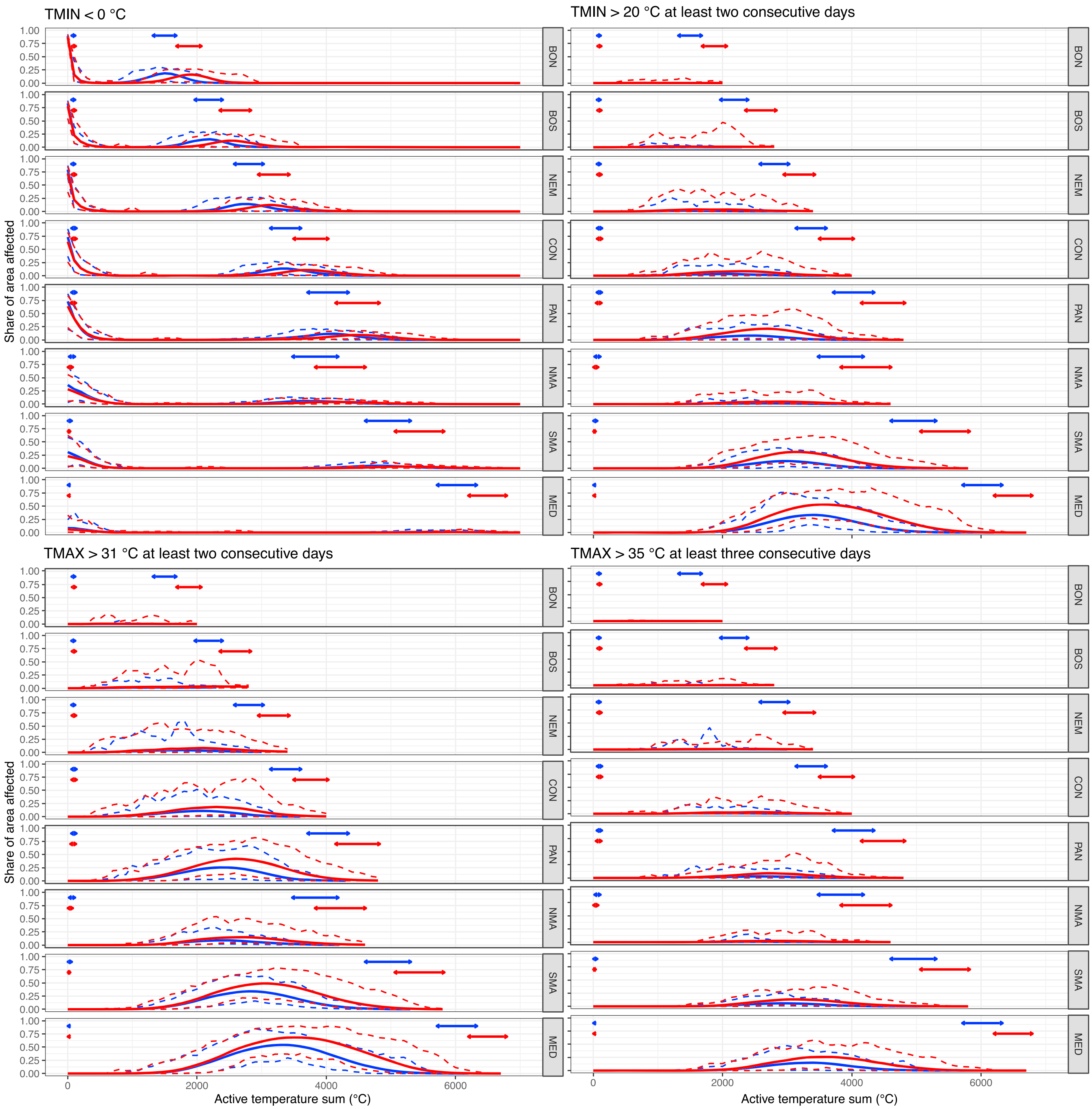

Figure 5. Relative area (fraction of the whole agro-climate zone) affected by climatic events based on four different air temperature thresholds for different agro-climate zones. Thick lines represent ensemble mean area affected as a function of active temperature sum, whereas lower and upper dashed lines represent the ensemble spread. Blue color refers to the baseline period between 1981 and 2010, whereas red color refers to model specific period defined by 2 ° C global warming (Table 1). The arrows represent a range of simulated dates for beginning of thermal growing season (first pair of arrows on *x* axis, typically corresponding to values of 200 DD) and the end of thermal growing season (second pair of arrows on *x* axis).

The area affected by frost generally decreases after the beginning of the growing season in the climates of western and southern Europe (Figure 5a). In contrast, the simulated area affected by frost after the start of growing season does not change substantially in the eastern and northern European agro-climate zones. In the southern boreal zone, there is even an increased risk of frost events with larger spatial extent compared to the baseline period, having implications for late spring frost (after the onset of vegetation). The area affected by first autumn (or winter) frost in eastern and northern European agro-climate zones occurs much later in terms of phenological timing under the 2 ° C warming scenario than during the baseline period, coinciding with an increase of *ATS* between 500 and 1,000 DD.

The climate change projections indicate a substantial increase in areas affected by warm temperature extremes. The share of area affected by mild heat stress is projected to increase over all of Europe; even in the northern Boreal climate zone, which did not experience these events during the baseline period, substantial areas will be affected. Mild heat stress events are also expected to occur earlier in the growing season in terms of phenological timing especially in eastern European agro-climate zones, increasing the risk for early heat stress impact on winter wheat. The maximum area affected by mild heat stress in the Pannonian, southern maritime, and Mediterranean zones is roughly doubled with respect to the baseline period, occurring nearly 500 DD later in terms of phenological timing (Figure 5).

The simulated patterns of change are similar for the area affected by strong heat stress. The most pronounced increase is expected in the Mediterranean agro-climate zone, with the area roughly doubling the baseline values. An increase in area affected is expected in all agro-climate zones, except in the northern boreal where it does not appear. Even though in terms of phenological timing, the first occurrence of events are not expected to change significantly, they might appear slightly earlier in the northern maritime, Mediterranean, and nemoral zones. The area affected by tropical nights is expected to increase in major parts of Europe, but most significantly in the Mediterranean, Pannonian, and southern maritime climates. Similar to the other two indicators of heat stress, the area affected roughly doubles in these regions under the 2 ° C warming scenario.

In Northern Europe the crop varieties with a long growing cycle will benefit most from the northward movement of climate zones. Nevertheless, the area affected by subzero temperatures after the beginning of thermal growing season is an important factor to consider here; while this area is expected to decrease in the maritime and the Mediterranean zones, no change or even moderate increase is expected in the eastern and northern European climate zones (Figure 5). Phenologically responsive species might therefore experience more frost damage in the future warming climate due to earlier onset of growing season (Ma et al., 2018). This conundrum could delay the expansion of agricultural production toward the North for several decades (Peltonen-Saino, 2009), until climate gets warm enough that the probability of late frost is very low.

Earlier sowing of crops to avoid heat stress and drought during the most sensitive period around flowering (Olesen et al., 2011) might increase the risk of late spring frost damage across northern and eastern European agro-climate zones. Simultaneously, mild heat stress is expected to occur earlier in phenological timing in climates of northern and eastern Europe. The temporal window of opportunity for favorable temperatures between last frost and first heat stress might therefore shrink, further challenging future adaptation options. Furthermore, intensification of summer heat stress is projected to affect a substantially higher portion of European areas (except boreal zones) in the future. Using longer crop varieties (i.e., needing a longer temperature accumulation), as an adaptation measure aiming to increase the grain filling period, might in reality be partly counterbalanced by detrimental effect due to the impacts of heat stress, especially in climates of southern, central, and eastern Europe. Lengthening of the thermal growing season due to earlier onset in spring and delay of ending in autumn in many European climates is likely to result in limited or no benefit for agricultural production due to excessive heat during the summer period as well as droughts. These agronomic constraints are indeed suggested by a number of modeling studies on winter wheat in Europe (Semenov & Shewry, 2011; Semenov et al., 2014; Trnka et al., 2015), indicating the need for improvement of heat stress tolerance in future wheat genotypes.

Our analysis suggests that heat stress will affect greater areas once the middle of the annual *ATS* accumulation has been reached. Maximum areas exposed to heat stress in the future will occur later in the phenological season when compared to the baseline period. Additionally, the duration of exposure to extreme heat will be substantially prolonged. Our study confirms that a longer exposure to heat stress will, besides drought (Trnka et al., 2015), be among the most important limiting factors for agricultural production in the Mediterranean, southern maritime, and Pannonian climates in the coming decades. Further, northward expansion of suitable areas for agricultural production in the future will be accompanied by a reduction of suitable areas in the Mediterranean due to temperature increase and excessive dryness (Fraga et al., 2012; Hannah et al., 2013). Although not the subject of this paper, the accelerating northward migration of agro-climatic zones might be accompanied with an increasing spread of pest species and diseases, and mounting severity and economic impacts of outbreaks (Baker et al., 2015).

While understanding the spatial patterns of migration of European agro-climate zones can offer an insight on possible future evolution and expansion of agriculture by understanding the production in the same climates during the baseline period, we demonstrated that careful analysis of the dynamical changes of exposure to extreme events during the critical periods of the growing season is necessary. While theoretically a lengthening of growing season and higher amounts of temperature accumulation in northern Europe would allow growing crop varieties that were typical for more southern agro-climate zones, an increase in intraseasonal and interannual temperature variability as well as other extreme climatic events (e.g. Peltonen-Saino, 2009) could also lead to higher risk for crop production losses, making the introduction of varieties with longer growth cycle less attractive.

5. Conclusions

This study demonstrates the migration velocity of agro-climate zones in Europe under past and future climate conditions. Using high-resolution gridded observational climate data, we calculate that Europe has already experienced a considerable migration of agro-climate zones toward the North over the last 40 years. The migration with velocities of up to 100 km per 10 years was the most pronounced in eastern and northeastern Europe. Western Europe has seen strong northward shift of maritime climate zones, appearing in regions such as central-northern Germany, substituting continental climate. The area affected by frost has decreased across Europe in last 40 years, most notably in eastern Europe in autumn and early winter. The area affected by mild and strong heat stress has increased especially in southern and eastern Europe. Observed changes have already shifted the timing of phenological phases of the main crops grown in Europe. Our analysis for winter wheat shows an earlier occurrence of flowering and maturity dates, however, with spatially varying implications for crop yields, depending on the coincidence of heat stress occurrence and sensitive crop growth stages. Overall, the Mediterranean, Pannonian, and southern maritime climates have experienced the most pronounced increase of area affected by heat stress, which has likely contributed to wheat yield stagnation in recent decades (Brisson et al., 2010).

The impact of global warming on agro-climate zone migration in Europe is assessed according to high-end emission scenario (RCP8.5) in our study. The velocity of climate zone migration depends on the chosen emission scenario, as the trends in surface warming reflect the transient nature of the scenarios (Bärring & Strandberg, 2018). Nevertheless, choosing RCP8.5 enables us to explore the consequences of the high emission trajectory.

We use high-resolution outputs of five regional climate models to appreciate a variety of regional climate changes corresponding to a 2 ° C global climate warming signal, typically reached in the next two to three decades according to RCP8.5. These projections suggest that *GSL* and *ATS* will on average increase by 19 days and 430 DD across Europe, with important regional differences. Further, northward migration of agro-climate zones is expected with an even accelerated pace in eastern and northeastern Europe, when compared to the observational period. While theoretically lengthening of growing season and higher *ATS* in northern Europe would allow growing crop varieties that were typical for agro-climate zones further south, increased interannual and intraseasonal temperature variability (including extremes) along with droughts could increase the risk of introducing crop varieties with longer growth cycle.

Using high-resolution climate scenarios, we calculate that northward climate zone migration will affect a large portion of Europe. For example, almost the entire region of southeastern Europe, currently characterized by a Pannonian climate, will beget southern maritime climate characteristics with substantially longer *GSL* and higher *ATS*. The area affected by heat stress will increase across major part of Europe, in addition to substantially prolonged duration of exposure to heat stress. Increased pressure due to heat stress, accompanied with water shortage (Trnka et al., 2015), will decrease suitability for crop growth especially during the middle and late summer in southern and southeastern Europe, which will challenge adaptation decisions favoring winter and (especially) summer crops with long crop cycle. The assessment of area affected by temperature extremes exposes another challenging factor related to shrinking of temporal window of opportunity for favorable temperatures between last frost and early heat stress, prominent especially in eastern and northern European climates. Spatial comparison of present-day farming systems to their future analogs and the exchange of knowledge between farming systems in different European agro-climate zones can suggest suitable adaptation strategies and technologies to be tested and validated (Kahiluoto et al., 2018).

While our study focuses on temperature-driven assessments of crop suitability in the presence of migrating agro-climate zones, drought has also been identified as a prominent crop growth limiting factor across most agro-climate zones (e.g., Olesen et al., 2011). Especially under longer growing season conditions, water stress will play an important role as a factor in shaping areas of future crop suitability. Drought can compromise the benefits of longer growing seasons in northern Europe and lead to crop yield failure in regions with limited water resources for irrigation. Adaptation of agriculture to climate change therefore needs to consider both factors to increase overall resilience.

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MCYFS gridded meteorological data can be retrieved online (at https://agri4cast.jrc.ec.europa.eu/DataPortal/Index.aspx). Bias-corrected EURO-CORDEX high-resolution regional climate model simulations can be obtained online (from https://cidportal.jrc.ec.europa.eu/ftp/jrc-opendata/LISCOAST/10011/LATEST). Wheat harvested areas are available at the Food and Agriculture Organization of the United Nations website (http://www.fao.org/faostat).

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